Greatest Common Divisor (GCD)

1. Find $\gcd(12, 18)$
2. Find $\gcd(24, 36)$
3. Find $\gcd(28, 42)$
4. Find $\gcd(45, 75)$
5. Find $\gcd(54, 90)$
6. Find $\gcd(64, 96)$
7. Find $\gcd(30, 45, 75)$
8. Find $\gcd(48, 72, 120)$
9. A florist has 36 red roses and 48 white roses. What is the greatest number of identical bouquets she can make using all the roses, and how many of each colour will be in each bouquet?
10. A rectangular courtyard measures 84 m by 108 m. It is to be paved with identical square slabs of the largest possible whole-metre side length without cutting any slabs. What is the side length of each slab, and how many slabs are needed in total?

Solutions

1. **6**
   Explanation: List all positive divisors of each number and identify the largest shared divisor:

$$\text{Divisors of } 12 : 1, 2, 3, 4, \mathbf{6}, 12$$
$$\text{Divisors of } 18 : 1, 2, 3, \mathbf{6}, 9, 18$$

   The greatest common divisor is 6.

2. **12**
   Explanation: List the positive divisors of 24 and 36:

$$\text{Divisors of } 24 : 1, 2, 3, 4, 6, 8, \mathbf{12}, 24$$
$$\text{Divisors of } 36 : 1, 2, 3, 4, 6, 9, \mathbf{12}, 18, 36$$

   Thus, $\gcd(24, 36) = 12$.

3. **14**
   Explanation: Using prime factorisation:

$$28 = 2^2 \times 7$$
$$42 = 2 \times 3 \times 7$$

   Multiply the lowest powers of the shared prime factors: $\gcd(28, 42) = 2 \times 7 = 14$.

4. **15**
   Explanation: Using prime factorisation:

$$45 = 3^2 \times 5$$
$$75 = 3 \times 5^2$$

   Take the lowest power of each shared prime factor: $\gcd(45, 75) = 3 \times 5 = 15$.

5. **18**
   Explanation: Using prime factorisation:

$$54 = 2 \times 3^3$$
$$90 = 2 \times 3^2 \times 5$$

   Multiply the lowest powers of the common prime factors: $\gcd(54, 90) = 2 \times 3^2 = 2 \times 9 = 18$.

6. **32**
   Explanation: Using prime factorisation:

$$64 = 2^6$$
$$96 = 2^5 \times 3$$

   The only shared prime factor is 2, with lowest power 2^5: $\gcd(64, 96) = 2^5 = 32$.

7. **15**
   Explanation: Find the prime factorisation of each of the three numbers:

$$30 = 2 \times 3 \times 5$$
$$45 = 3^2 \times 5$$
$$75 = 3 \times 5^2$$

   Taking the lowest power of each prime common to all three numbers gives $\gcd(30, 45, 75) = 3 \times 5 = 15$.

8. **24**
   Explanation: Find the prime factorisation of each number:

$$48 = 2^4 \times 3$$
$$72 = 2^3 \times 3^2$$
$$120 = 2^3 \times 3 \times 5$$

   Taking the lowest power of each shared prime factor gives $\gcd(48, 72, 120) = 2^3 \times 3 = 8 \times 3 = 24$.

9. **12 bouquets (3 red roses and 4 white roses per bouquet)**
   Explanation: The greatest number of identical bouquets is given by $\gcd(36, 48)$:

$$36 = 2^2 \times 3^2$$
$$48 = 2^4 \times 3$$

   Thus, $\gcd(36, 48) = 2^2 \times 3 = 12$ bouquets. Each bouquet will contain $\dfrac{36}{12} = 3$ red roses and $\dfrac{48}{12} = 4$ white roses.

10. **Side length of 12 m, requiring 63 slabs in total**
    Explanation: The largest possible square slab side length that divides both dimensions evenly is $\gcd(84, 108)$:

$$84 = 2^2 \times 3 \times 7$$
$$108 = 2^2 \times 3^3$$

    Thus, $\gcd(84, 108) = 2^2 \times 3 = 12$ m. Along the 84 m side, $\dfrac{84}{12} = 7$ slabs fit; along the 108 m side, $\dfrac{108}{12} = 9$ slabs fit. In total, $7 \times 9 = 63$ slabs are needed.